

6.1.3 Charging and discharging capacitors

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) (i) charging and discharging capacitor through a resistor	
(ii) techniques and procedures to investigate the charge and the discharge of a capacitor using both meters and data-loggers	PAG9 HSW4 Investigating the charge and discharge of capacitors in the laboratory.
(b) time constant of a capacitor–resistor circuit; $\tau = CR$	
(c) equations of the form $x = x_0 e^{-\frac{t}{CR}}$ and $x = x_0 (1 - e^{-\frac{t}{CR}})$ for capacitor–resistor circuits	Learners will be expected to know how $\ln x-t$ graphs can be used to determine CR . <i>M0.5, M2.5, M3.10, M3.12</i>
(d) graphical methods and spreadsheet modelling of the equation $\frac{\Delta Q}{\Delta t} = -\frac{Q}{CR}$ for a discharging capacitor	HSW3 Using spreadsheets to model the discharge of a capacitor. <i>M3.9</i>
(e) exponential decay graph; constant-ratio property of such a graph.	<i>M3.11</i>